



(Don't Write anything in this Area)

- 3.b) While genomic research offers immense promise for personalized healthcare, it also raises significant ethical concerns. Critically examine the ethical implications of using genomic data for promoting human welfare." (15 Marks)

Genomic research refers to the study, evaluation and analysis of the human genome (entire 3.1 billion base pairs)

The promise of personalised health care by genomic research

① It helps in understanding the individuals unique genome

↳ helps in Pharmacogenomics.

↳ helps in the treatment of Tumour and the Cancer cells.

Recently, in the USA, a women was cured from [liver cancer] using personalised genome analysis.

② It helps in understanding the evolutionary pattern of the various genes.

(egs) Pabbo has found out various genes of Neanderthals is present in European population.

→ The issue of late waking up and sleeping sickness is attributed to this.

③ helps in Catering the endemic and population disease control.

(egs) vaccines of COVID developed due to genomic research.

④ It further helps in criminal investigation → mental trauma and any kind of health issues can be analysed via the genomic research.

Ethical implications of using Genomic data :-

- ① Issue of Confidentiality :- The biological data are used without the consent.
 - ② Issue of using the data as making clones, designer babies.
(egs) China is accused of doing it.
 - ③ It generates power equation between the researcher and subject.
 - ④ It also pose threat to country's sovereignty. (egs) many research, testing are done on developing country.
 - ⑤ It poses threat to life of the human and their dignity.
- Hence, it is seen that balanced approach is what needed in the genomic research.